

INTRODUCTION TO BIOSTATISTICS

DEFINITIONS, SCOPE AND LIMITATIONS

Introduction

In the modern world of computers and information technology, the importance of statistics is very well recognised by all the disciplines. Statistics has originated as a science of statehood and found applications slowly and steadily in Agriculture, Economics, Commerce, Biology, Medicine, Industry, planning, education and so on. As on date there is no other human walk of life, where statistics cannot be applied.

Biostatistics

- ✓ The science of collecting, analyzing, and interpreting biological or health data using statistical methods
- ✓ It is critical for evidence-based nursing practice, public health initiatives, and improving patient outcomes.
- ✓ Biostatistics is an essential science that integrates knowledge from statistics, computer science, and life sciences to analyze health and biological data. It involves designing research studies, collecting and organizing data, summarizing findings, and analyzing results to reach reliable conclusions. The role of biostatistics has greatly expanded due to increased data availability and the growing need for skilled biostatisticians.
- ✓ A key concept in biostatistics is understanding populations and samples. The "population" is the complete group we wish to study—for example, all children under five years in a community when examining disease prevalence. Often, it's impractical to study the entire population due to time, cost, and resources. Therefore, we use a smaller, representative subset called a "sample."

Two fundamental goals when working with samples are:

1. Summarizing the key characteristics of the population.
 2. Ensuring that findings from the sample have enough statistical support to generalize to the entire population.
- ✓ Biostatistics helps identify relationships between risk factors and health outcomes. For instance, when studying diabetes in a community, rather than testing everyone, we collect data from a sample. From this sample, we estimate the disease prevalence and identify risk factors associated with diabetes.

Biostatistics involves two main branches:

1. Descriptive Statistics:

- △ It focuses on organizing and summarizing data in a clear, understandable way. Examples include calculating averages, creating frequency tables, and generating charts to visualize data clearly.
- △ Summarizes data to describe patterns and trends.
 - Measures of Central Tendency: Mean, median, mode
 - Measures of Dispersion: Range, variance, standard deviation

- Graphical representations like histograms and pie charts are used for visualization

2. Inferential Statistics:

△ It uses data from samples to make decisions or predictions about larger populations. For example, using a sample to estimate how widespread a disease is in the overall population.

△ Draws conclusions about populations based on sample data.

- Hypothesis Testing: Evaluating the null hypothesis versus the alternative hypothesis
- Confidence Intervals: Estimating population parameters with a range of values
- p-value Interpretation: Assessing statistical significance

A biostatistician's role includes:

- Designing and conducting studies effectively.
- Understanding appropriate data collection methods.
- Knowing the strengths and limitations of various statistical methods.
- Applying statistical techniques correctly to interpret health-related data.

Why biostatistics in nursing

For nurses, understanding these basic concepts helps in:

1. Ultimately, biostatistics provides the tools to make informed, evidence-based decisions in health sciences.
2. Understanding biostatistics helps nurses and healthcare professionals improve patient care by.
 - interpreting research findings or clinical studies to inform practice.
 - identifying health trends and making effective clinical decisions.
3. Evaluating patient outcomes and health interventions.
 - Conducting or participating in research aimed at improving patient care and health services.

Role of Biostatistics in real-life applications

Application Area	Key Contributions
Clinical Trials	Ensures valid study designs, randomization, sample size optimization, data analysis.
Epidemiology	Identifies disease patterns, risk factors, intervention impacts.
Genetics/Genomics	Maps genes, studies inheritance patterns, analyzes genetic variability.
Public Health Policy	Evaluates interventions, guides resource allocation, addresses health disparities.
Drug Development	Tests drug efficacy/safety, predicts dosing regimens, analyzes pharmacokinetics.
Environmental Health	Studies links between environmental exposures and health outcomes.
Evidence-Based Medicine	Assesses scientific evidence strength for medical decisions/policies.
Quality Improvement	Tracks healthcare performance metrics to improve patient outcomes.
Bioinformatics	Analyzes large-scale biological data for personalized medicine/drug discovery.
Demography	Studies population dynamics for healthcare planning/adjustments.

Rare Diseases Research	Designs trials with limited samples; subgroup analyses for personalized treatments.
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Statistics

Statistics involves collecting, organizing, summarizing, analyzing data, and making decisions based on these data. The primary goal is to use a representative part (sample) to make accurate conclusions about a larger group (population). Understanding this process is critical in nursing, as it allows informed decision-making in clinical practice, public health, and research.

Origin and Growth of Statistics:

The word 'Statistics' and 'Statistical' are all derived from the Latin word Status, means a political state. The theory of statistics as a distinct branch of scientific method is of comparatively recent growth. Research particularly into the mathematical theory of statistics is rapidly proceeding and fresh discoveries are being made all over the world.

Meaning of Statistics:

Statistics is concerned with scientific methods for collecting, organising, summarising, presenting and analysing data as well as deriving valid conclusions and making reasonable decisions based on this analysis. Statistics is concerned with the systematic collection of numerical data and its interpretation.

The word 'statistic' is used to refer to

1. Numerical facts, such as the number of people living in particular area.
2. The study of ways of collecting, analysing and interpreting the facts.

Definitions:

Statistics is defined differently by different authors over a period of time. In the olden days statistics was confined to only state affairs but in modern days it embraces almost every sphere of human activity. Therefore, a number of old definitions, which was confined to narrow field of enquiry were replaced by more definitions, which are much more comprehensive and exhaustive. Secondly, statistics has been defined in two different ways – Statistical data and statistical methods.

The following are some of the definitions of statistics as numerical data.

1. Statistics are the classified facts representing the conditions of people in a state. In particular they are the facts, which can be stated in numbers or in tables of numbers or in any tabular or classified arrangement.
2. Statistics are measurements, enumerations or estimates of natural phenomenon usually systematically arranged, analysed and presented as to exhibit important interrelationships among them.

Definitions by A.L. Bowley:

Statistics are numerical statement of facts in any department of enquiry placed in relation to each other. - A.L. Bowley Statistics may be called the science of counting in one of the departments due to Bowley, obviously this is an incomplete definition as it takes into account

only the aspect of collection and ignores other aspects such as analysis, presentation and interpretation.

Bowley gives another definition for statistics, which states ‘statistics may be rightly called the scheme of averages. This definition is also incomplete, as averages play an important role in understanding and comparing data and statistics provide more measures.

Definition by Croxton and Cowden:

Statistics may be defined as the science of collection, presentation analysis and interpretation of numerical data from the logical analysis. The definition of statistics by Croxton and Cowden is the most scientific and realistic one.

According to this definition there are four stages:

1. Collection of Data:

It is the first step, and this is the foundation upon which the entire data set. Careful planning is essential before collecting the data. There are different methods of collection of data such as census, sampling, primary, secondary, etc., and the investigator should make use of correct method.

2. Presentation of data:

The mass data collected should be presented in a suitable, concise form for further analysis. The collected data may be presented in the form of tabular or diagrammatic or graphic form.

3. Analysis of data:

The data presented should be carefully analysed for making inference from the presented data such as measures of central tendencies, dispersion, correlation, regression etc.,

4. Interpretation of data:

The final step is drawing conclusion from the data collected. A valid conclusion must be drawn on the basis of analysis. A high degree of skill and experience is necessary for the interpretation.

Definition by Horace Secrist:

Statistics may be defined as the aggregate of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated according to a reasonable standard of accuracy, collected in a systematic manner, for a predetermined purpose and placed in relation to each other.

The above definition seems to be the most comprehensive and exhaustive.

Populations can be:

- Finite (countable): like all infants born in a hospital during a specific year.
- Infinite (endless): like all possible blood sugar levels a diabetic patient could theoretically have over time.

Population Size (N)

Population size refers to the total number of elements or individuals within a defined population. For example, if there are 100 infants under five years in a community, the population size (N) is 100.

Sample

A sample is a smaller group selected from a larger population intended to represent the entire population. Because studying every individual in a population is often impractical, samples are used to infer characteristics about the whole group.

For example, rather than assessing every child under five in a large community to find disease prevalence, researchers may select a sample of 30 children to represent the entire group.

Sample Size (n)

The sample size is the number of elements included in a sample, represented by the lowercase letter "n". If a sample contains 30 children, then the sample size (n) is 30.

Parameter and Statistic

- **Parameter:** This is a measurable characteristic describing the entire population. For instance, the actual average weight of all diabetic patients in a hospital is a parameter.
- **Statistic:** This represents the characteristic measured from a sample. For example, the average weight measured from 30 diabetic patients randomly selected from a hospital is a statistic. Statistics are used to estimate parameters.

Probability and Probability Distributions

- Understanding the likelihood of events occurring.
 - **Discrete Distributions:** Binomial and Poisson distributions.
 - **Continuous Distributions:** Normal distribution, which is widely applied in healthcare research.

Sampling Techniques

- Methods to select representative subsets from a population:
 - Probability Sampling (e.g., random sampling).
 - Non-probability Sampling (e.g., convenience sampling)
- Proper sampling ensures unbiased and reliable results.

Statistical Modelling

- Techniques like correlation and regression analysis to examine relationships between variables.
 - Simple linear regression predicts outcomes based on one independent variable.
 - Correlation coefficients measure the strength of relationships.

Vital Statistics

- Statistical measures related to birth rates, mortality rates, morbidity rates, and disease prevalence.
- These are used to assess population health and healthcare effectiveness.

Functions of Statistics

There are many functions of statistics. Let us consider the following five important functions.

1. **Condensation:**

Generally speaking by the word 'to condense', we mean to reduce or to lessen. Condensation is mainly applied at embracing the understanding of a huge mass of data by providing only few observations. If in a particular class in Chennai School, only marks in an examination are given, no purpose will be served. Instead, if we are given the average mark in that particular examination, definitely it serves the better purpose.

Similarly, the range of marks is also another measure of the data. Thus, Statistical measures help to reduce the complexity of the data and consequently to understand any huge mass of data.

2. **Comparison:**

Classification and tabulation are the two methods that are used to condense the data. They help us to compare data collected from different sources. Grand totals, measures of central tendency measures of dispersion, graphs and diagrams, coefficient of correlation etc provide ample scope for comparison.

If we have one group of data, we can compare within itself. If the rice production (in Tonnes) in Tanjore district is known, then we can compare one region with another region within the district. Or if the rice production (in Tonnes) of two different districts within Tamilnadu is known, then also a comparative study can be made. As statistics is an aggregate of facts and figures, comparison is always possible and in fact comparison helps us to understand the data in a better way.

3. **Forecasting:**

By the word forecasting, we mean to predict or to estimate before hand. Given the data of the last ten years connected to rainfall of a particular district in Tamilnadu, it is possible to predict or forecast the rainfall for the near future. In business also forecasting plays a dominant role in connection with production, sales, profits etc. The analysis of time series and regression analysis plays an important role in forecasting.

4. **Estimation:**

One of the main objectives of statistics is drawn inference about a population from the analysis for the sample drawn from that population. The four major branches of statistical inference are

- a) Estimation theory
- b) Tests of Hypothesis
- c) Non-Parametric tests
- d) Sequential analysis

In estimation theory, we estimate the unknown value of the population parameter based on the sample observations. Suppose we are given a sample of heights of hundred students in

a school, based upon the heights of these 100 students, it is possible to estimate the average height of all students in that school.

5. Tests of Hypothesis:

A statistical hypothesis is some statement about the probability distribution, characterising a population on the basis of the information available from the sample observations. In the formulation and testing of hypothesis, statistical methods are extremely useful. Whether crop yield has increased because of the use of new fertilizer or whether the new medicine is effective in eliminating a particular disease are some examples of statements of hypothesis and these are tested by proper statistical tools.

Scope of Statistics

Statistics is not a mere device for collecting numerical data, but as a means of developing sound techniques for their handling, analysing and drawing valid inferences from them. Statistics is applied in every sphere of human activity – social as well as physical – like Biology, Commerce, Education, Planning, Business Management, Information Technology, etc. It is almost impossible to find a single department of human activity where statistics cannot be applied. We now discuss briefly the applications of statistics in other disciplines.

i. Statistics and Industry:

Statistics is widely used in many industries. In industries, control charts are widely used to maintain a certain quality level. In production engineering, to find whether the product is conforming to specifications or not, statistical tools, namely inspection plans, control charts, etc., are of extreme importance. In inspection plans we have to resort to some kind of sampling – a very important aspect of Statistics.

ii. Statistics and Commerce:

Statistics are lifeblood of successful commerce. Any businessman cannot afford to either by under stocking or having overstock of his goods. In the beginning he estimates the demand for his goods and then takes steps to adjust with his output or purchases. Thus, statistics is indispensable in business and commerce.

As so many multinational companies have invaded into our Indian economy, the size and volume of business is increasing. On one side the stiff competition is increasing whereas on the other side the tastes are changing, and new fashions are emerging. In this connection, market survey plays an important role to exhibit the present conditions and to forecast the likely changes in future.

iii. Statistics and Agriculture:

Analysis of variance (ANOVA) is one of the statistical tools developed by Professor R.A. Fisher, plays a prominent role in agriculture experiments. In tests of significance based on small samples, it can be shown that statistics is adequate to test the significant difference between two sample means. In analysis of variance, we are concerned with the testing of equality of several population means.

For an example, five fertilizers are applied to five plots each of wheat and the yield of wheat on each of the plots are given. In such a situation, we are interested in finding out whether the effect of these fertilisers on the yield is significantly different or not. In other words, whether the samples are drawn from the same normal population or not. The answer to this problem is provided by the technique of ANOVA and it is used to test the homogeneity of several population means.

iv. **Statistics and Economics:**

Statistical methods are useful in measuring numerical changes in complex groups and interpreting collective phenomenon. Nowadays the uses of statistics are abundantly made in any economic study. Both in economic theory and practice, statistical methods play an important role.

Alfred Marshall said, “Statistics are the straw only which I like every other economist have to make the bricks”. It may also be noted that statistical data and techniques of statistical tools are immensely useful in solving many economic problems such as wages, prices, production, distribution of income and wealth and so on. Statistical tools like Index numbers, time series Analysis, Estimation theory, Testing Statistical Hypothesis are extensively used in economics.

v. **Statistics and Education:**

Statistics is widely used in education. Research has become a common feature in all branches of activities.

Statistics is necessary for the formulation of policies to start new course, consideration of facilities available for new courses etc. There are many people engaged in research work to test the past knowledge and evolve new knowledge. These are possible only through statistics.

vi. **Statistics and Planning:**

Statistics is indispensable in planning. In the modern world, which can be termed as the “world of planning”, almost all the organisations in the government are seeking the help of planning for efficient working, for the formulation of policy decisions and execution of the same.

In order to achieve the above goals, the statistical data relating to production, consumption, demand, supply, prices, investments, income expenditure etc and various advanced statistical techniques for processing, analysing and interpreting such complex data are of importance. In India statistics play an important role in planning, commissioning both at the central and state government levels.

vii. **Statistics and Medicine:**

In Medical sciences, statistical tools are widely used. In order to test the efficiency of a new drug or medicine, t - test is used or to compare the efficiency of two drugs or two medicines, ttest for the two samples is used. More and more applications of statistics are at present used in clinical investigation.

viii. **Statistics and Modern applications:**

Recent developments in the fields of computer technology and information technology have enabled statistics to integrate their models and thus make statistics a part of decision-making procedures of many organisations. There are so many software packages available for solving design of experiments, forecasting simulation problems etc.

SYSTAT, a software package offers more scientific and technical graphing options than any other desktop statistics package. SYSTAT supports all types of scientific and technical research in various diversified fields as follows

- a) **Archaeology:** Evolution of skull dimensions
- b) **Epidemiology:** Tuberculosis
- c) **Statistics:** Theoretical distributions
- d) **Manufacturing:** Quality improvement
- e) **Medical research:** Clinical investigations.
- f) **Geology:** Estimation of Uranium reserves from ground water

Limitations of statistics

Statistics with all its wide application in every sphere of human activity has its own limitations. Some of them are given below.

1. Statistics is not suitable to the study of qualitative phenomenon

Since statistics is basically a science and deals with a set of numerical data, it is applicable to the study of only those subjects of enquiry, which can be expressed in terms of quantitative measurements.

As a matter of fact, qualitative phenomenon like honesty, poverty, beauty, intelligence etc, cannot be expressed numerically and any statistical analysis cannot be directly applied on these qualitative phenomena.

Nevertheless, statistical techniques may be applied indirectly by first reducing the qualitative expressions to accurate quantitative terms. For example, the intelligence of a group of students can be studied on the basis of their marks in a particular examination.

2. Statistics does not study individuals

Statistics does not give any specific importance to the individual items, in fact it deals with an aggregate of objects. Individual items, when they are taken individually do not constitute any statistical data and do not serve any purpose for any statistical enquiry.

3. Statistical laws are not exact

It is well known that mathematical and physical sciences are exact. But statistical laws are not exact, and statistical laws are only approximations. Statistical conclusions are not universally true. They are true only on an average.

4. Statistics table may be misused

Statistics must be used only by experts; otherwise, statistical methods are the most dangerous tools in the hands of the inexperienced. The use of statistical tools by the inexperienced and untrained persons might lead to wrong conclusions. Statistics can be easily misused by quoting wrong figures of data. As King says aptly 'statistics are like clay of which one can make a God or Devil as one pleases'.

5. Statistics is only, one of the methods of studying a problem:

Statistical method do not provide complete solution of the problems because problems are to be studied taking the background of the countries culture, philosophy or religion into consideration. Thus, the statistical study should be supplemented by other evidences.

INTRODUCTION TO SAMPLING METHODS

Introduction

Sampling is very often used in our daily life. For example, while purchasing food grains from a shop we usually examine a handful from the bag to assess the quality of the commodity. A doctor examines a few drops of blood as sample and draws conclusion about the blood constitution of the whole body. Thus, most of our investigations are based on samples. In this chapter, let us see the importance of sampling and the various methods of sample selections from the population.

Population

In a statistical enquiry, all the items, which fall within the purview of enquiry, are known as **Population** or **Universe**. In other words, the population is a complete set of all possible observations of the type which is to be investigated. Total number of students studying in a school or college, total number of books in a library, total number of houses in a village or town are some examples of population.

Sometimes it is possible and practical to examine every person or item in the population we wish to describe. We call this a **Complete enumeration**, or **census**. We use **sampling** when it is not possible to measure every item in the population. Statisticians use the word population to refer not only to people but to all items that have been chosen for study.

Finite population and infinite population

A population is said to be finite if it consists of finite number of units. Number of workers in a factory, production of articles in a particular day for a company are examples of finite population. The total number of units in a population is called population size. A population is said to be infinite if it has infinite number of units. For example, the number of stars in the sky, the number of people seeing the Television programmes etc.,

Census Method

Information on population can be collected in two ways – census method and sample method. In census method every element of the population is included in the investigation. For example, if we study the average annual income of the families of a particular village or area, and if there are 1000 families in that area, we must study the income of all 1000 families. In this method no family is left out, as each family is a unit.

Merits and limitations of Census method:

Merits:

1. The data are collected from each and every item of the population
2. The results are more accurate and reliable, because every item of the universe is required.
3. Intensive study is possible.
4. The data collected may be used for various surveys, analyses etc.

Limitations:

1. It requires a large number of enumerators, and it is a costly method
2. It requires more money, labour, time energy etc.
3. It is not possible in some circumstances where the universe is infinite.

Sampling

The theory of sampling has been developed recently but this is not new. In our everyday life we have been using sampling theory as we have discussed in introduction. In all those cases we believe that the samples give a correct idea about the population. Most of our decisions are based on the examination of a few items that is sample studies.

Sample

Statisticians use the word **sample** to describe a portion chosen from the population. A finite subset of statistical individuals defined in a population is called a sample. The number of units in a sample is called the **sample size**.

Sampling unit

The constituents of a population which are individuals to be sampled from the population and cannot be further subdivided for the purpose of the sampling at a time are called sampling units. For example, to know the average income per family, the head of the family is a sampling unit. To know the average yield of rice, each farm owner's yield of rice is a sampling unit.

Sampling frame

For adopting any sampling procedure, it is essential to have a list identifying each sampling unit by a number. Such a list or map is called sampling frame. A list of voters, a list of house holders, a list of villages in a district, a list of farmers etc. are a few examples of sampling frame.

Reasons for selecting a sample

Sampling is inevitable in the following situations:

1. Complete enumerations are practically impossible when the population is infinite.
2. When the results are required in a short time.
3. When the area of survey is wide.
4. When resources for survey are limited particularly in respect of money and trained persons.
5. When the item or unit is destroyed under investigation.

Parameters and statistics

We can describe samples and populations by using measures such as the mean, median, mode and standard deviation. When these terms describe the characteristics of a population, they are called **parameters**. When they describe the characteristics of a sample, they are called **statistics**. A parameter is a characteristic of a population, and a statistic is a characteristic of a sample. Since samples are subsets of population statistics provide estimates of the parameters. That is, when the parameters are unknown, they are estimated from the values of the statistics.

In general, we use Greek or capital letters for population parameters and lower-case Roman letters to denote sample statistics. N , μ , σ , are the standard symbols for the size, mean,

S.D, of population. n , $\bar{x}$, s , are the standard symbol for the size, mean, s.d of sample respectively].

Principles of Sampling:

Samples have to provide good estimates. The following principle tell us that the sample methods provide such good estimates

1. Principle of statistical regularity:

A moderately large number of units chosen at random from a large group are almost sure on the average to possess the characteristics of the large group.

2. Principle of Inertia of large numbers:

Other things being equal, as the sample size increases, the results tend to be more accurate and reliable.

3. Principle of Validity:

This states that the sampling methods provide valid estimates about the population units (parameters).

4. Principle of Optimisation:

This principle takes into account the desirability of obtaining a sampling design which gives optimum results. This minimizes the risk or loss of the sampling design.

The foremost purpose of sampling is to gather maximum information about the population under consideration at minimum cost, time and human power. This is best achieved when the sample contains all the properties of the population.

Sampling errors and non-sampling errors:

The two types of errors in a sample survey are sampling errors and non - sampling errors.

1. Sampling errors:

Although a sample is a part of population, it cannot be expected generally to supply full information about population. So, there may be in most cases difference between statistics and parameters. The discrepancy between a parameter and its estimate due to sampling process is known as **sampling error**.

2. Non-sampling errors:

In all surveys some errors may occur during collection of actual information. These errors are called **non-sampling errors**.

Advantages of Sampling

There are many advantages of sampling methods over census method. They are as follows:

1. Sampling saves time and labour.
2. It results in reduction of cost in terms of money and man hour.
3. Sampling ends up with greater accuracy of results.
4. It has greater scope

5. It has greater adaptability.
6. If the population is too large, or hypothetical or destroyable sampling is the only method to be used.

Limitation of Sampling

The limitations of sampling are given below:

1. Sampling is to be done by qualified and experienced persons. Otherwise, the information will be unbelievable.
2. Sample method may give the extreme values sometimes instead of the mixed values.
3. There is the possibility of sampling errors. Census survey is free from sampling error.

Types of Sampling

The technique of selecting a sample is of fundamental importance in sampling theory and it depends upon the nature of investigation.

The sampling procedures which are commonly used may be classified as

1. Probability sampling.
2. Non-probability sampling.
3. Mixed sampling.

1. Probability sampling (Random sampling):

A probability sample is one where the selection of units from the population is made according to known probabilities. (eg.) Simple random sample, probability proportional to sample size etc.

2. Non-Probability sampling:

It is the one where discretion is used to select 'representative' units from the population (or) to infer that a sample is 'representative' of the population. This method is called **judgement or purposive** sampling. This method is mainly used for opinion surveys; A common type of judgement sample used in surveys is quota sample.

This method is not used in general because of prejudice and bias of the enumerator. However, if the enumerator is experienced and expert, this method may yield valuable results. For example, in the market research survey of the performance of their new car, the sample was all new car purchasers.

3. Mixed Sampling:

Here samples are selected partly according to some probability and partly according to a fixed sampling rule; they are termed as mixed samples and the technique of selecting such samples is known as **mixed sampling**.

Methods of selection of samples

Here we shall consider the following three methods:

1. Simple random sampling.
2. Stratified random sampling.
3. Systematic random sampling.

1. Simple Random Sampling

A simple random sample from finite population is a sample selected such that each possible sample combination has equal probability of being chosen. It is also called unrestricted random sampling.

1. Simple random sampling without replacement:

In this method the population elements can enter the sample only once (ie) the units once selected is not returned to the population before the next draw.

2. Simple random sampling with replacement:

In this method the population units may enter the sample more than once. Simple random sampling may be with or without replacement.

Methods of selection of a simple random sampling:

The following are some methods of selection of a simple random sampling.

a) Lottery Method:

This is the most popular and simplest method. In this method all the items of the population are numbered on separate slips of paper of same size, shape and colour. They are folded and mixed up in a container. The required numbers of slips are selected at random for the desired sample size. For example, if we want to select 5 students, out of 50 students, then we must write their names or their roll numbers of all the 50 students on slips and mix them. Then we make a random selection of 5 students.

This method is mostly used in lottery draws. If the universe is infinite this method is inapplicable.

b) Table of Random numbers:

As the lottery method cannot be used, when the population is infinite, the alternative method is that of using the table of random numbers. There are several standard tables of random numbers.

1. Tippett's table
2. Fisher and Yates' table
3. Kendall and Smith's table are the three tables among them.

A random number table is so constructed that all digits 0 to 9 appear independent of each other with equal frequency. If we have to select a sample from population of size $N=100$, then the numbers can be combined three by three to give the numbers from 001 to 100.

Procedure to select a sample using random number table:

Units of the population from which a sample is required are assigned with equal number of digits. When the size of the population is less than thousand, three digit number 000, 001, 002, 999 are assigned. We may start at any place and may go on in any direction such as column wise or row-wise in a random number table. But consecutive numbers are to be used.

On the basis of the size of the population and the random number table available with us, we proceed according to our convenience. If any random number is greater than the population size N , then N can be subtracted from the random number drawn. This can be repeatedly until the number is less than N or equal to N .

Example 1:

In an area there are 500 families. Using the following extract from a table of random numbers select a sample of 15 families to find out the standard of living of those families in that area.

4652	3819	8431	2150	2352	2472	0043	3488
9031	7617	1220	4129	7148	1943	4890	1749
2030	2327	7353	6007	9410	9179	2722	8445
0641	1489	0828	0385	8488	0422	7209	4950

Solution:

In the above random number table, we can start from any row or column and read three digit numbers continuously row-wise or column wise.

Now we start from the third row, the numbers are:

203	023	277	353	600	794	109	179
272	284	450	641	148	908	280	

Since some numbers are greater than 500, we subtract 500 from those numbers and we rewrite the selected numbers as follows:

203	023	277	353	100	294	109	179	272	284	450	141
148	408	280									

c) Random number selections using calculators or computers

Random number can be generated through scientific calculator or computers. For each press of the key get a new random number. The ways of selection of sample is similar to that of using random number table.

Merits of using random numbers

Merits:

1. Personal bias is eliminated as a selection depends solely on chance.
2. A random sample is in general a representative sample for a homogenous population.
3. There is no need for the thorough knowledge of the units of the population.
4. The accuracy of a sample can be tested by examining another sample from the same universe when the universe is unknown.
5. This method is also used in other methods of sampling.

Limitations:

1. Preparing lots or using random number tables is tedious when the population is large.
2. When there is large difference between the units of population, the simple random sampling may not be a representative sample.
3. The size of the sample required under this method is more than that required by stratified random sampling.

4. It is generally seen that the units of a simple random sample lie apart geographically. The cost and time of collection of data are more.

3. Stratified Random Sampling

Of all the methods of sampling the procedure commonly used in surveys is stratified sampling. This technique is mainly used to reduce the population heterogeneity and to increase the efficiency of the estimates. Stratification means division into groups.

In this method the population is divided into a number of subgroups or strata. The strata should be so formed that each stratum is homogeneous as far as possible. Then from each stratum a simple random sample may be selected, and these are combined together to form the required sample from the population.

Types of Stratified Sampling

There are two types of stratified sampling.

They are **proportional** and **non-proportional**.

In the **proportional sampling** equal and proportionate representation is given to subgroups or strata. If the number of items is large, the sample will have a higher size and vice versa.

The population size is denoted by N and the sample size is denoted by ' n ' the sample size is allocated to each stratum in such a way that the sample fractions is a constant for each stratum. That is given by $n/N = c$. So, in this method each stratum is represented according to its size.

In non-proportionate sample, equal representation is given to all the sub-strata regardless of their existence in the population.

Example 2:

A sample of 50 students is to be drawn from a population consisting of 500 students belonging to two institutions A and B. The number of students in the institution A is 200 and the institution B is 300. How will you draw the sample using proportional allocation?

Solution:

There are two strata in this case with sizes $N_1 = 200$ and $N_2 = 300$ and the total population $N = N_1 + N_2 = 500$ The sample size is 50.

If n_1 and n_2 are the sample sizes,

$$n_1 = \frac{n}{N} \times N_1 = \frac{50}{500} \times 200 = 20$$

$$n_2 = \frac{n}{N} \times N_2 = \frac{50}{500} \times 300 = 30$$

The sample sizes are 20 from A and 30 from B. Then the units from each institution are to be selected by simple random sampling.

Merits and limitations of stratified sampling:

Merits:

1. It is more representative.
2. It ensures greater accuracy.
3. It is easy to administer as the universe is sub - divided.
4. Greater geographical concentration reduces time and expenses.
5. When the original population is badly skewed, this method is appropriate.
6. For non – homogeneous population, it may field good results.

Limitations:

1. To divide the population into homogeneous strata, it requires more money, time and statistical experience which is a difficult one.
2. Improper stratification leads to bias, if the different strata overlap such a sample will not be a representative one.

Systematic Sampling

This method is widely employed because of its ease and convenience. A frequently used method of sampling when a complete list of the population is available is **systematic sampling**. It is also called **Quasi-random sampling**.

Selection procedure:

The whole sample selection is based on just a random start. The first unit is selected with the help of random numbers and the rest get selected automatically according to some pre-designed pattern is known as **systematic sampling**.

With systematic random sampling every K^{th} element in the frame is selected for the sample, with the starting point among the first K elements determined at random.

For example, if we want to select a sample of 50 students from 500 students under this method K^{th} item is picked up from the sampling frame and K is called the **sampling interval**.

$$\text{Sampling interval, } K = \frac{N}{n} = \frac{\text{Population Size}}{\text{Sample Size}}$$

$$K = \frac{50}{500} = 10$$

$K = 10$ is the sampling interval. Systematic sample consists in selecting a random number say i K and every K^{th} unit subsequently. Suppose the random number 'i' is 5, then we select 5, 15, 25, 35, 45,..... The random number 'i' is called random start. The technique will generate K systematic samples with equal probability.

Merits:

1. This method is simple and convenient.

2. Time and work is reduced much.
3. If proper care is taken result will be accurate.
4. It can be used in infinite population.

Limitations:

1. Systematic sampling may not represent the whole population.
 2. There is a chance of personal bias of the investigators.
- Systematic sampling is preferably used when the information is to be collected from trees in a forest, house in blocks, entries in a register which are in a serial order etc.

COLLECTION OF DATA, CLASSIFICATION AND TABULATION

Introduction

Everybody collects, interprets and uses information, much of it in a numerical or statistical forms in day-to-day life. It is a common practice that people receive large quantities of information everyday through conversations, televisions, computers, the radios, newspapers, posters, notices and instructions. It is just because there is so much information available that people need to be able to absorb, select and reject it. In everyday life, in business and industry, certain statistical information is necessary, and it is independent to know where to find it how to collect it. As consequences, everybody has to compare prices and quality before making any decision about what goods to buy. As employees of any firm, people want to compare their salaries and working conditions, promotion opportunities and so on. In time the firms on their part want to control costs and expand their profits.

One of the main functions of statistics is to provide information which will help on making decisions. Statistics provides the type of information by providing a description of the present, a profile of the past and an estimate of the future. The following are some of the objectives of collecting statistical information.

1. To describe the methods of collecting primary statistical information.
2. To consider the status involved in carrying out a survey.
3. To analyse the process involved in observation and interpreting.
4. To define and describe sampling.
5. To analyse the basis of sampling.
6. To describe a variety of sampling methods.

Data

Data are the raw materials of statistics. Data collection methods include primary data (newly collected through surveys or experiments) and secondary data (already collected from previous studies, hospital records, or databases).

When presenting data, we use visual tools like bar charts for qualitative data, histograms, and frequency polygons for quantitative data. These visuals help in quickly understanding data distribution patterns, such as which categories or values are most common.

Types/ category of Data

Data used in biostatistics can be either quantitative (numerical, such as weight or blood pressure) or qualitative (categorical, such as gender or blood type).

1. **Quantitative Data (Numerical Data):** Data represented by numbers. Numerical variables that can be discrete (e.g., number of patients) or continuous (e.g., blood pressure readings).
Quantitative data can be:
 - **Discrete:** Countable values (e.g., number of hospital visits, number of children in a family).
 - **Continuous:** Measurable values that can take any value within a range (e.g., weight, blood sugar level, height).
2. **Qualitative Data (Categorical Data):** Data categorized based on attributes or characteristics. Categorical variables that represent characteristics (e.g., gender, blood type) These can include:
 - **Nominal:** Categories without a specific order (e.g., blood type, gender).
 - **Ordinal:** Categories with a clear order (e.g., pain levels: mild, moderate, severe).

Understanding data types is essential for selecting appropriate statistical methods.

Sources of Data

Data come from two main sources:

- **Primary Data:** Collected directly through new surveys, experiments, or observations. This includes using questionnaires, interviews, physical measurements, or clinical assessments.
- **Secondary Data:** Data previously collected for another purpose, such as hospital records, public health databases, or published research findings.

Scales /Levels of Measurement

Measurement levels help determine how data can be analysed. There are four main levels:

1. **Nominal Scale:**
 - Only categorizes or labels data. Categories without a specific order
 - No inherent order, distance, or true zero point.
 - Example: Blood groups /types (A, B, AB, O).
2. **Ordinal Scale:**
 - Categorizes data with a natural order. Ordered categories
 - Differences between categories are not necessarily equal or meaningful.
 - Example: Pain severity (mild, moderate, severe).
3. **Interval Scale:**

- Categories with ordered data and meaningful equal distances between values. Numeric values with equal intervals but no true zero point, so ratios are not meaningful.
- Example: Temperature in Celsius. Zero degrees Celsius does not mean absence of temperature.

4. Ratio Scale:

- Highest measurement level. Numeric values with a true zero (e.g., weight, height). Data has order, meaningful distances, and a true zero point.
- Example: Weight, height, age. (e.g., 60 kg is twice as heavy as 30 kg, and zero kg means no weight).

Statistical investigation is a comprehensive and requires systematic collection of data about some group of people or objects, describing and organizing the data, analyzing the data with the help of different statistical method, summarizing the analysis and using these results for making judgements, decisions and predictions. The validity and accuracy of final judgement is most crucial and depends heavily on how well the data was collected in the first place. The quality of data will greatly affect the conditions and hence at most importance must be given to this process and every possible precaution should be taken to ensure accuracy while collecting the data.

Nature of data:

It may be noted that different types of data can be collected for different purposes. The data can be collected in connection with time or geographical location or in connection with time and location.

The following are the three types of data:

1. Time series data.
2. Spatial data
3. Spacio-temporal data.

Time series data:

It is a collection of a set of numerical values, collected over a period of time. The data might have been collected either at regular intervals of time or irregular intervals of time.

Example 1:

The following is the data for the three types of expenditures in rupees for a family for the four years 2001,2002,2003,2004.

Year	Food	Education	Others	Total
2001	3000	2000	3000	8000
2002	3500	3000	4000	10500
2003	4000	3500	5000	12500
2004	5000	5000	6000	16000

Spatial Data:

If the data collected is connected with that of a place, then it is termed as spatial data. For example, the data may be

- i. Number of runs scored by a batsman in different test matches in a test series at different places
- ii. District wise rainfall in Freetown
- iii. Prices of silver in four metropolitan cities

Example 2:

The population of the southern states of India in 1991.

State	Population
Kailahun	5,56,38,318
Freetown	6,63,04,854
Kenema	4,48,17,398
Bo	2,90,11,237
Koinadugu	7,89,416

Spacio Temporal Data:

If the data collected is connected to the time as well as place then it is known as spacio temporal data.

Example 3:

State	Population	
	2007	2020
Kailahun	4,82,97,456	5,56,38,318
Freetown	5,34,03,619	6,63,04,854
Kenema	3,70,43,451	4,48,17,398
Koinadugu	2,54,03,217	2,90,11,237
Bonthe	6,04,136	7,89,416

Categories of data

Any statistical data can be classified under two categories depending upon the sources utilized.

These categories are,

1. Primary data
2. Secondary data

Primary data:

Primary data is the one, which is collected by the investigator himself for the purpose of a specific inquiry or study. Such data is original in character and is generated by survey conducted by individuals or research institution or any organisation.

Example 4:

If a researcher is interested to know the impact of noonmeal scheme for the school children, he has to undertake a survey and collect data on the opinion of parents and children by asking relevant questions. Such a data collected for the purpose is called primary data.

The primary data can be collected by the following five methods.

- a) Direct personal interviews.
- b) Indirect Oral interviews.
- c) Information from correspondents.
- d) Mailed questionnaire method.
- e) Schedules sent through enumerators.

Direct personal interviews:

The persons from whom information are collected are known as informants. The investigator personally meets them and asks questions to gather the necessary information. It is the suitable method for intensive rather than extensive field surveys. It suits best for intensive study of the limited field.

Merits:

- i People willingly supply information because they are approached personally. Hence, more response noticed in this method than in any other method.
- ii The collected information are likely to be uniform and accurate. The investigator is there to clear the doubts of the informants.
- iii Supplementary information on informant's personal aspects can be noted. Information on character and environment may help later to interpret some of the results.
- iv Answers for questions about which the informant is likely to be sensitive can be gathered by this method.
- v The wordings in one or more questions can be altered to suit any informant. Explanations may be given in other languages also. Inconvenience and misinterpretations are thereby avoided.

Limitations:

- i It is very costly and time consuming.
- ii It is very difficult, when the number of persons to be interviewed is large and the persons are spread over a wide area.
- iii Personal prejudice and bias are greater under this method.

Indirect Oral Interviews:

Under this method the investigator contacts witnesses or neighbours or friends or some other third parties who can supply the necessary information. This method is preferred if the required information is on addiction or cause of fire or theft or murder etc., If a fire has broken out a certain place, the persons living in neighbourhood and witnesses are likely to give information on the cause of fire. In some cases, police interrogated third parties who are supposed to have knowledge of a theft or a murder and get some clues. Enquiry committees appointed by governments generally adopt this method and get people's views and all possible details of facts relating to the enquiry. This method is suitable whenever direct sources do not exist or cannot be relied upon or would be unwilling to part with the information.

The validity of the results depends upon a few factors, such as the nature of the person whose evidence is being recorded, the ability of the interviewer to draw out information

from the third parties by means of appropriate questions and cross examinations, and the number of persons interviewed. For the success of this method one person or one group alone should not be relied upon.

Information from correspondents:

The investigator appoints local agents or correspondents in different places and compiles the information sent by them. Information to Newspapers and some departments of Government come by this method. The advantage of this method is that it is cheap and appropriate for extensive investigations. But it may not ensure accurate results because the correspondents are likely to be negligent, prejudiced and biased. This method is adopted in those cases where information is to be collected periodically from a wide area for a long time.

Mailed questionnaire method:

Under this method a list of questions is prepared and is sent to all the informants by post. The list of questions is technically called questionnaire. A covering letter accompanying the questionnaire explains the purpose of the investigation and the importance of correct information and request the informants to fill in the blank spaces provided and to return the form within a specified time. This method is appropriate in those cases where the informants are literates and are spread over a wide area.

Merits:

- i It is relatively cheap.
- ii It is preferable when the informants are spread over the wide area.

Limitations:

- i The greatest limitation is that the informants should be literates who are able to understand and reply to the questions.
- ii It is possible that some of the persons who receive the questionnaires do not return them.
- iii It is difficult to verify the correctness of the information furnished by the respondents.

With the view of minimizing non-respondents and collecting correct information, the questionnaire should be carefully drafted. There is no hard and fast rule. But the following general principles may be helpful in framing the questionnaire. A covering letter and a self addressed and stamped envelope should accompany the questionnaire. The covering letter should politely point out the purpose of the survey and privilege of the respondent who is one among the few associated with the investigation. It should assure that the information would be kept confidential and would never be misused. It may promise a copy of the findings or free gifts or concessions etc.,

Characteristics of a good questionnaire:

1. Number of questions should be minimum.
2. Questions should be in logical orders, moving from easy to more difficult questions.
3. Questions should be short and simple. Technical terms and vague expressions capable of different interpretations should be avoided.

4. Questions fetching YES or NO answers are preferable. There may be some multiple-choice questions requiring lengthy answers are to be avoided.
5. Personal questions and questions which require memory power and calculations should also be avoided.
6. Question should enable cross check. Deliberate or unconscious mistakes can be detected to an extent.
7. Questions should be carefully framed to cover the entire scope of the survey.
8. The wording of the questions should be proper without hurting the feelings or arousing resentment.
9. As far as possible confidential information should not be sought.
10. Physical appearance should be attractive, sufficient space should be provided for answering each question.

Schedules sent through Enumerators:

Under this method enumerators or interviewers take the schedules, meet the informants and filling their replies. Often distinction is made between the schedule and a questionnaire. A schedule is filled by the interviewers in a face-to-face situation with the informant. A questionnaire is filled by the informant which he receives and returns by post. It is suitable for extensive surveys.

Merits:

- i It can be adopted even if the informants are illiterates.
- ii Answers for questions of personal and pecuniary nature can be collected.
- iii Non-response is minimum as enumerators go personally and contact the informants.
- iv The information collected are reliable. The enumerators can be properly trained for the same.
- v It is most popular methods.

Limitations:

- i It is the costliest method.
- ii Extensive training is to be given to the enumerators for collecting correct and uniform information.
- iii Interviewing requires experience. Unskilled investigators are likely to fail in their work.

Before the actual survey, a pilot survey is conducted. The questionnaire/Schedule is pre-tested in a pilot survey. A few among the people from whom actual information is needed are asked to reply. If they misunderstand a question or find it difficult to answer or do not like its wordings etc., it is to be altered. Further it is to be ensured that every question fetches the desired answer.

Merits and Demerits of primary data:

1. The collection of data by the method of personal survey is possible only if the area covered by the investigator is small. Collection of data by sending the enumerator is bound to be expensive. Care should be taken twice that the enumerator records correct information provided by the informants.
2. Collection of primary data by framing a schedule or distributing and collecting questionnaires by post is less expensive and can be completed in shorter time.

3. Suppose the questions are embarrassing or of complicated nature or the questions probe into personnel affairs of individuals, then the schedules may not be filled with accurate and correct information and hence this method is unsuitable.
4. The information collected for primary data is more reliable than those collected from the secondary data.

Secondary Data

Secondary data are those data which have been already collected and analysed by some earlier agency for its own use; and later the same data are used by a different agency. According to W.A. Neiswanger, 'A primary source is a publication in which the data are published by the same authority which gathered and analysed them. A secondary source is a publication, reporting the data which have been gathered by other authorities and for which others are responsible'.

Sources of Secondary data

In most of the studies the investigator finds it impracticable to collect first-hand information on all related issues and as such he makes use of the data collected by others. There is a vast amount of published information from which statistical studies may be made and fresh statistics are constantly in a state of production.

The sources of secondary data can broadly be classified under two heads:

1. Published sources, and
2. Unpublished sources.

Published Sources:

The various sources of published data are:

- a) Reports and official publications of
 - i International bodies such as the International Monetary Fund, International Finance Corporation and United Nations Organisation.
 - ii Central and State Governments such as MoH Reports
- b) Semi-official publication of various local bodies such as Municipal Corporations and District Boards.
- c) Private publications-such as the publications of –
 - i Trade and professional bodies such as the Federation of Sierra Leone Chambers of Commerce and Institute of Chartered Accountants.
 - ii Financial and economic journals such as 'Commerce', 'Capital' and 'Sierra Leone Finance'.
 - iii Annual reports of joint stock companies.
 - iv Publications brought out by research agencies, research scholars, etc.

It should be noted that the publications mentioned above vary about the periodically of publication. Some are published at regular intervals (yearly, monthly, weekly etc.,) whereas others are ad hoc publications, i.e., with no regularity about periodicity of publications.

Note: A lot of secondary data is available on the internet. We can access it at any time for the further studies.

Unpublished Sources

All statistical material is not always published. There are various sources of unpublished data such as records maintained by various Government and private offices, studies made by research institutions, scholars, etc. Such sources can also be used where necessary

Precautions in the use of Secondary data

The following are some of the points that are to be considered in the use of secondary data

- 1 How the data has been collected and processed
- 2 The accuracy of the data
- 3 How far the data has been summarized
- 4 How comparable the data is with other tabulations
- 5 How to interpret the data, especially when figures collected for one purpose is used for another

With secondary data, people have to compromise between what they want and what they are able to find.

Merits and Demerits of Secondary Data:

1. Secondary data is cheap to obtain. Many government publications are relatively cheap and libraries stock quantities of secondary data produced by the government, by companies and other organisations.
2. Large quantities of secondary data can be got through internet.
3. Much of the secondary data available has been collected for many years and therefore it can be used to plot trends.
4. Secondary data is of value to:
 - The government – help in making decisions and planning future policy.
 - Business and industry – in areas such as marketing, and sales in order to appreciate the general economic and social conditions and to provide information on competitors.
 - Research organisations – by providing social, economical and industrial information.

Data Classification

The collected data, also known as raw data or ungrouped data are always in an unorganised form and need to be organised and presented in meaningful and readily comprehensible form in order to facilitate further statistical analysis. It is, therefore, essential for an investigator to condense a mass of data into more and more comprehensible and assimilable form. The process of grouping into different classes or sub classes according to some characteristics is known as classification, tabulation is concerned with the systematic arrangement and presentation of classified data. Thus, classification is the first step in tabulation.

For Example, letters in the post office are classified according to their destinations viz., USA, Canada, UK, etc.,

Objectives of Classification:

The following are main objectives of classifying the data:

1. It condenses the mass of data in an easily assimilable form.
2. It eliminates unnecessary details.
3. It facilitates comparison and highlights the significant aspect of data.
4. It enables one to get a mental picture of the information and helps in drawing inferences.
5. It helps in the statistical treatment of the information collected.

Types of classification:

Statistical data are classified in respect of their characteristics.

Broadly there are four basic types of classification namely

- A) Chronological classification
- B) Geographical classification
- C) Qualitative classification
- D) Quantitative classification

Chronological classification:

In chronological classification the collected data are arranged according to the order of time expressed in years, months, weeks, etc., The data is generally classified in ascending order of time. For example, the data related with population, sales of a firm, imports and exports of a country are always subjected to chronological classification.

Example 5:

The estimates of birth rates in India during 1970 – 76 are

Year	1970	1971	1972	1973	1974	1975	1976
Birth Rate	36.8	36.9	36.6	34.6	34.5	35.2	34.2

Geographical classification:

In this type of classification, the data are classified according to geographical region or place. For instance, the production of paddy in different states in India, production of wheat in different countries etc.,

Example 6:

Country	America	China	Denmark	France	India
Yield of wheat in (kg/acre)	1925	893	225	439	862

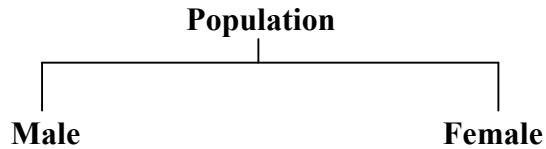
Qualitative classification:

In this type of classification data are classified on the basis of same attributes or quality like sex, literacy, religion, employment etc., Such attributes cannot be measured along with a scale.

For example, if the population to be classified in respect to one attribute, say sex, then we can classify them into two namely that of males and females. Similarly, they can also be classified into 'employed' or 'unemployed' on the basis of another attribute 'employment'.

Thus, when the classification is done with respect to one attribute, which is dichotomous in nature, two classes are formed, one possessing the attribute and the other not possessing the attribute. This type of classification is called simple or dichotomous classification.

A simple classification may be shown as under

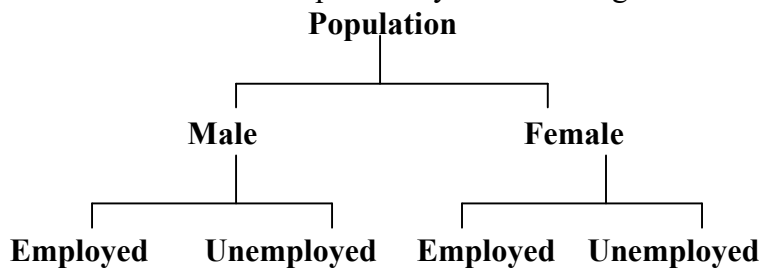


The classification, where two or more attributes are considered and several classes are formed, is called a manifold classification. For example, if we classify population simultaneously with respect to two attributes, e.g. sex and employment, then population are first classified with respect to 'sex' into 'males' and 'females'. Each of these classes may then be further classified into 'employment' and 'unemployment' based on attribute 'employment' and as such

Population is classified into four classes namely.

- (i) Male employed
- (ii) Male unemployed
- (iii) Female employed
- (iv) Female unemployed

Still the classification may be further extended by considering other attributes like marital status etc. This can be explained by the following chart



Quantitative classification:

Quantitative classification refers to the classification of data according to some characteristics that can be measured such as height, weight, etc., For example the students of a college may be classified according to weight as given below.

Weight (in lbs)	No of Students
90-100	50
100-110	200
110-120	260
120-130	360
130-140	90
140-150	40
Total	1000

In this type of classification there are two elements, namely

- (i) the variable (i.e.) the weight in the above example, and
- (ii) the frequency in the number of students in each class.

There are 50 students having weights ranging from 90 to 100 lb, 200 students having weight ranging between 100 to 110 lb and so on.

Tabulation:

Tabulation is the process of summarizing classified or grouped data in the form of a table so that it is easily understood, and an investigator is quickly able to locate the desired information. A table is a systematic arrangement of classified data in columns and rows. Thus, a statistical table makes it possible for the investigator to present a huge mass of data in a detailed and orderly form. It facilitates comparison and often reveals certain patterns in data which are otherwise not obvious. Classification and 'Tabulation', as a matter of fact, are not two distinct processes. They go together. Before tabulation data are classified and then displayed under different columns and rows of a table.

Advantages of Tabulation

Statistical data arranged in a tabular form serve following objectives:

1. It simplifies complex data, and the data presented are easily understood.
2. It facilitates comparison of related facts.
3. It facilitates computation of various statistical measures like averages, dispersion, correlation etc.
4. It presents facts in minimum possible space and unnecessary repetitions and explanations are avoided. Moreover, the needed information can be easily located.
5. Tabulated data are good for references, and they make it easier to present the information in the form of graphs and diagrams.

Preparing a Table:

The making of a compact table itself an art. This should contain all the information needed within the smallest possible space. What the purpose of tabulation is and how the tabulated information is to be used are the main points to be kept in mind while preparing for a statistical table.

An ideal table should consist of the following main parts:

1. Table number
2. Title of the table
3. Captions or column headings
4. Stubs or row designation
5. Body of the table
6. Footnotes
7. Sources of data

Table Number

A table should be numbered for easy reference and identification. This number, if possible, should be written in the centre at the top of the table. Sometimes it is also written just before the title of the table.

Title

A good table should have a clearly worded, brief but unambiguous title explaining the nature of data contained in the table. It should also state arrangement of data, and the period covered. The title should be placed centrally on the top of a table just below the table number (or just after table number in the same line).

Captions or column Headings

Captions in a table stands for brief and self-explanatory headings of vertical columns. Captions may involve headings and sub-headings as well. The unit of data contained should also be given for each column. Usually, a relatively less important and shorter classification should be tabulated in the columns.

Stubs or Row Designations

Stubs stand for brief and self-explanatory headings of horizontal rows. Normally, a relatively more important classification is given in rows. Also, a variable with a large number of classes is usually represented in rows. For example, rows may stand for score of classes and columns for data related to sex of students. In the process, there will be many rows for scores classes but only two columns for male and female students.

A model structure of a table is given below:

Table Number		Title of the Table	
Sub Heading	Caption Headings		Total
	Caption Sub-Headings		
Stub Sub- Headings	Body		
Total			

Foot notes:

Sources Note:

Body

The body of the table contains the numerical information of frequency of observations in the different cells. This arrangement of data is according to the description of captions and stubs.

Footnotes:

Footnotes are given at the foot of the table for explanation of any fact or information included in the table which needs some explanation. Thus, they are meant for explaining or providing further details about the data, that have not been covered in title, captions and stubs.

Sources of data:

Lastly one should also mention the source of information from which data are taken. This may preferably include the name of the author, volume, page and the year of publication. This should also state whether the data contained in the table is of 'primary or secondary' nature.

Requirements of a Good Table:

A good statistical table is not merely a careless grouping of columns and rows but should be such that it summarizes the total information in an easily accessible form in minimum possible space. Thus, while preparing a table, one must have a clear idea of the information to be presented, the facts to be compared and the points to be stressed.

Though, there is no hard and fast rule for forming a table, yet a few general points should be kept in mind:

1. A table should be formed in keeping with the objects of statistical enquiry.
2. A table should be carefully prepared so that it is easily understandable.
3. A table should be formed so as to suit the size of the paper. But such an adjustment should not be at the cost of legibility.
4. If the figures in the table are large, they should be suitably rounded or approximated. The method of approximation and units of measurements too should be specified.
5. Rows and columns in a table should be numbered and certain figures to be stressed may be put in 'box' or 'circle' or in bold letters.
6. The arrangements of rows and columns should be in a logical and systematic order. This arrangement may be alphabetical, chronological or according to size.
7. The rows and columns are separated by single, double or thick lines to represent various classes and sub-classes used. The corresponding proportions or percentages should be given in adjoining rows and columns to enable comparison. A vertical expansion of the table is generally more convenient than the horizontal one.
8. The averages or totals of different rows should be given at the right of the table and that of columns at the bottom of the table. Totals for every sub-class too should be
9. mentioned.
10. In case it is not possible to accommodate all the information in a single table, it is better to have two or more related tables.

Type of Tables

Tables can be classified according to their purpose, stage of enquiry, nature of data or number of characteristics used. On the basis of the number of characteristics, tables may be classified as follows:

1. Simple or one-way table
2. Two-way table
3. Manifold table

Simple or one-way Table:

A simple or one-way table is the simplest table which contains data of one characteristic only. A simple table is easy to construct and simple to follow. For example, the blank table given below may be used to show the number of adults in different occupations in a locality.

The number of adults in different occupations in a locality

Occupations	No. Of Adults
Total	

Two-way Table:

A table, which contains data on two characteristics, is called a two-way table. In such case, therefore, either stub or caption is divided into two co-ordinate parts. In the given table, as an example the caption may be further divided in respect of 'sex'. This subdivision is shown in two-way table, which now contains two characteristics namely, occupation and sex.

The number of adults in a locality in respect of occupation and sex

Occupation	No. of Adults		Total
	Male	Female	
Total			

Manifold Table:

Thus, more and more complex tables can be formed by including other characteristics. For example, we may further classify the caption sub-headings in the above table in respect of "marital status", "religion" and "socio-economic status" etc. A table, which has more than two characteristics of data is considered as a manifold table. For instance, table shown below shows three characteristics namely, occupation, sex and marital status.

Occupation			No. of Adults			Total
		Male		Female		
	M	U	Total	M	U	Total
Total						

Foot note: M Stands for Married and U stands for unmarried.

Manifold tables, though complex are good in practice as these enable full information to be incorporated and facilitate analysis of all related facts. Still, as a normal practice, not more than four characteristics should be represented in one table to avoid confusion. Other related tables may be formed to show the remaining characteristics

VARIABLES USED IN BIOSTATISTICS

Variables and Random Variables

- Variable: A characteristic or attribute measured or observed in data collection.
- Random Variable: A variable whose values result from random events (e.g., selecting patients randomly for a study).

The most common types of variables used in biostatistics can be broadly categorized into **quantitative variables** and **qualitative variables**, with further subtypes under each category.

1. Quantitative Variables

Quantitative variables represent numerical data that can be measured or counted.

- **Discrete Variables:**
 - These are countable variables that take specific, distinct values.
 - Examples: Number of patients in a hospital, number of medications prescribed.
- **Continuous Variables:**
 - These can take any value within a range and are often measured rather than counted.
 - Examples: Blood pressure, body temperature, height, weight.

2. Qualitative Variables

Qualitative variables, also known as categorical variables, represent characteristics or categories that do not have numerical values.

- **Nominal Variables:**
 - These are categories without any intrinsic order.
 - Examples: Blood type (A, B, AB, O), gender (male, female).
- **Ordinal Variables:**
 - These categories have a meaningful order or ranking but the intervals between them are not equal.
 - Examples: Pain levels (mild, moderate, severe), education levels (high school, bachelor's degree, master's degree).
- **Binary (Dichotomous) Variables:**
 - A special type of nominal variable with only two possible outcomes.
 - Examples: Yes/No responses (e.g., presence or absence of a disease), alive/dead status.

3. Independent and Dependent Variables

These variables are commonly used in experimental and observational studies:

- **Independent Variable:**
 - The variable manipulated or categorized to observe its effect on another variable.
 - Example: Dosage of medication administered.

- **Dependent Variable:**

- The outcome variable that is measured in response to changes in the independent variable.
- Example: Reduction in blood pressure after medication.

4. Composite Variables

These are created by combining multiple variables into a single measure for complex analyses.

- Example: Overall health score derived from factors like genetics, medical care, and lifestyle behaviours.

5. Confounding Variables

These are extraneous variables that may affect the relationship between independent and dependent variables if not controlled.

- Example: Age or comorbidities affecting treatment outcomes.

Types of Random Variables

1. Quantitative Random Variables:

- **Discrete:** Take specific, countable values (e.g., number of medications a patient takes).
- **Continuous:** Can take any value within a given range (e.g., blood pressure levels).

2. Qualitative Random Variables:

- **Nominal:** Categories without order (e.g., gender, nationality).
- **Ordinal:** Categories with meaningful order (e.g., education levels: primary, secondary, college).

Summary Table

Type	Subtype	Examples
Quantitative	Discrete	Number of patients, number of medications
	Continuous	Blood pressure, height, weight
Qualitative	Nominal	Blood type, gender
	Ordinal	Pain levels, education levels
Experimental Design	Binary/Dichotomous	Yes/No responses, alive/dead
	Independent Variable	Medication dosage
	Dependent Variable	Blood pressure reduction
Derived Variables	Composite	Overall health score